

Machine Learning (CAI-262)

Objectives:

- Performance Evaluation
 - Accuracy
 - Confusion Matrix
 - ROC and PR Curves
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Performance Evaluation:

While building any machine learning model, the first thing that comes to our mind is how we can build an accurate and good model. Model performance in general refers to how well a model accomplishes its intended task. Performance evaluation is the quantitative measure of how well a trained model performs on specific model evaluation metrics. This information can then be used to determine if a model is ready to move onto the next stage of testing, be deployed more broadly, or is in need of more training or retraining. There are multiple performance metrics available to assess the performance of classification and regression tasks. We will discuss only classification metrics in detail given below.

1) Accuracy:

Accuracy is a metric that measures how often a machine learning model correctly predicts the outcome. You can calculate accuracy by dividing the number of correct predictions by the total number of predictions. You can measure the accuracy on a scale of 0 to 1 or as a percentage. The higher the accuracy, the better. You can achieve a perfect accuracy of 1.0 when every prediction the model makes is correct.

$$\text{Accuracy} = \frac{\text{Correct predictions}}{\text{All predictions}}$$

Drawback of Accuracy:

Accuracy is a helpful metric when you deal with balanced classes and care about the overall model “correctness” and not the ability to predict a specific class.

Suppose that you are working in a company, and you are asked to create a model that, based on various measurements, predicts whether a product is defective or not. You decide to use your favorite classifier, train it on the data and you get a 96.2% accuracy.

Your boss is astonished and decides to use your model without any further tests. A few weeks later he enters your office and underlines the uselessness of your model. Indeed, the model you created has not found any defective product from the time it has been used in production.

After some investigations, you find out that there is only around 3.8% of the product made by your company that are defective and your model just always answers “not defective”, leading to a 96.2% accuracy. These results you obtained is due to the imbalanced dataset you are working with, so there are different methods that can be used to tackle classification problems with imbalanced classes.

2) Confusion Matrix:

In binary classification, there are two possible target classes, which are typically labeled as "positive" and "negative" or "1" and "0". In our spam example above, the target (positive class) is "spam," and the negative class is "not spam."

When evaluating the accuracy, we looked at correct and wrong predictions disregarding the class label. However, in binary classification, we can be "correct" and "wrong" in two different ways.

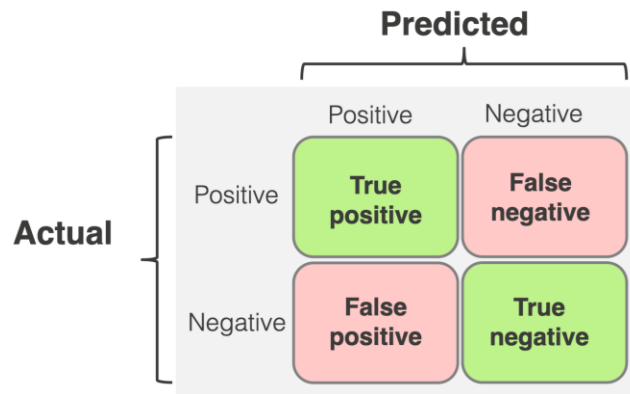
Correct predictions include true positives and true negatives.

- **True positive (TP):** A sample is actually positive and is correctly classified by the model as positive.
- **True negative (TN):** An sample is actually negative and is correctly classified by the model as negative.

Model errors include false positives and false negatives.

- **False Positive (FP):** A sample is actually negative but is incorrectly classified by the model as positive ("false alarm").
- **False Negative (FN):** A sample is actually positive but is incorrectly classified by the model as negative.

Using the confusion matrix, you can visualize all 4 different outcomes in a single table as follows.



To evaluate how well the model deals with identifying and predicting True Positives, we should measure precision and recall instead of accuracy.

Precision is a metric that measures how often a machine learning model correctly predicts the positive class. You can calculate precision by dividing the number of correct positive predictions (true positives) by the total number of instances the model predicted as positive (both true and false positives). In other words, precision answers the question: how often the positive predictions are correct?

$$Precision = \frac{TP}{TP + FP}$$

Recall is a metric that measures how often a machine learning model correctly identifies positive instances (true positives) from all the actual positive samples in the dataset. In other words, recall answers the question: can an ML model find all instances of the positive class?

$$Recall = \frac{TP}{TP + FN}$$

Precision Recall trade off:

Some of the models in machine learning require more precision and some models require more recall. So, it is important to know the balance between Precision and recall or, simply, precision-recall trade-off.

F1 score is calculated as the harmonic mean of the precision and recall scores, as shown below. It ranges from 0-100%, and a higher F1 score denotes a better-quality classifier.

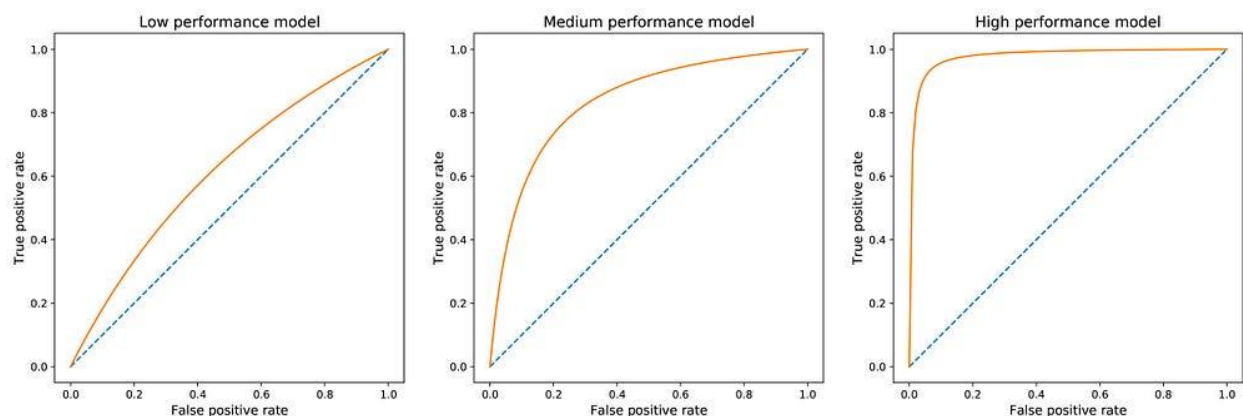
$$\begin{aligned} \text{F1 Score} &= \frac{2}{\frac{1}{\text{Precision}} + \frac{1}{\text{Recall}}} \\ &= \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \end{aligned}$$

3) ROC and PR Curves

Another interesting metric is the **ROC curve** (standing for Receiver Operating Characteristic), defined with respect to a given class (that we will denote C in the following).

Suppose that for a given point x , we have a model that outputs the probability that this point belongs to C: $P(C | x)$. Based on this probability, we can define a decision rule that consists in saying that x belongs to class C if and only if $P(C | x) \geq T$, where T is a given threshold defining our decision rule. If $T=1$, a point is labelled as belonging to C only if the model is 100% confident it does. If $T=0$, every points are labelled as belonging to C.

Each value of the threshold T generates a point (false positive, true positive) and, then, the ROC curve is the curve described by the ensemble of points generated when T varies from 1 to 0. This curve starts at point (0,0), ends at point (1,1) and is increasing. A good model will have a curve that increases quickly from 0 to 1 (meaning that only a little precision has to be sacrificed to get a high recall).

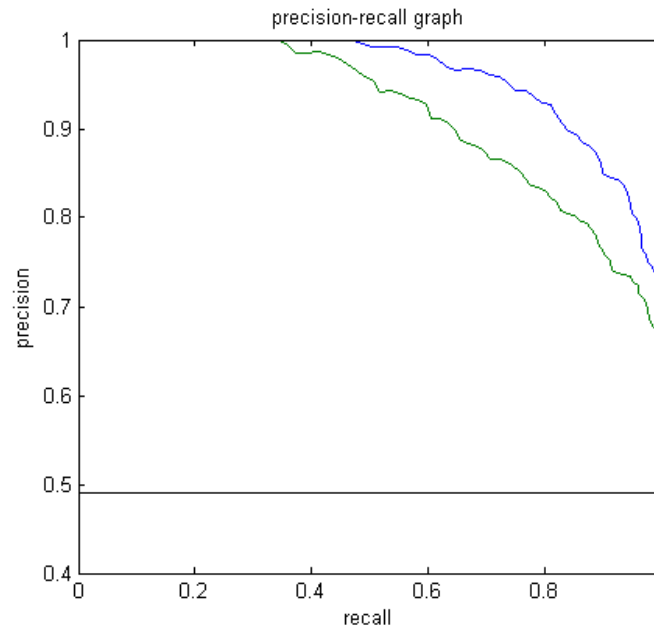


A **PR curve** is simply a graph with Precision values on the y-axis and Recall values on the x-axis.

It is important to note that Precision is also called the Positive Predictive Value (PPV).

The recall is also called Sensitivity, Hit Rate, or True Positive Rate (TPR).

The figure below shows a comparison of sample PR curves.



It is desired that the algorithm should have both high precision and high recall. However, most machine learning algorithms often involve a trade-off between the two. A good PR curve has greater AUC (area under the curve). In the figure above, the classifier corresponding to the blue line has better performance than the classifier corresponding to the green line.
